

Whisper_Architecture

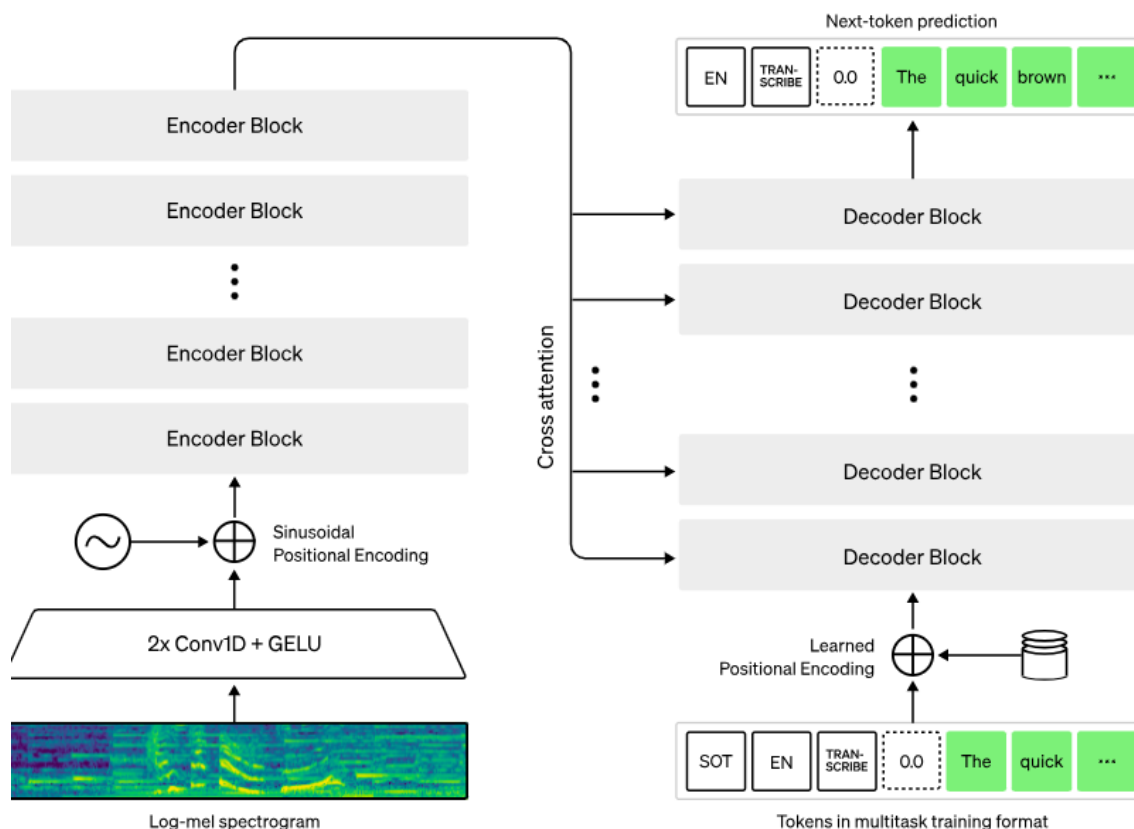


Figure 1 Whisper_Architecture

1. Log-Mel Spectrogram-Raw audio is not directly given to the Transformer. First, audio is converted into a **Log-Mel Spectrogram**.
2. 2× Conv1D + GELU-This means Whisper uses **two 1D convolution layers** before the Transformer encoder.
3. Sinusoidal Positional Encoding-Transformer models do not naturally know the order of input frames. So positional encoding tells the model:

This sound came first,
this sound came next,

4. Encoder Blocks-The encoder reads the Log-Mel Spectrogram and converts it into a high-level audio representation.

5. Cross Attention-This connects the encoder and decoder.The decoder looks at the encoded audio information while generating text.
6. Tokens in Multitask Training Format-Whisper does not output full words at once. It outputs tokens one by one.
7. Learned Positional Encoding in Decoder-the decoder uses positional encoding to understand token order.

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8. Decoder Blocks-The decoder generates the transcript one token at a time.

It uses:

1. Previous generated tokens.
2. Encoder audio representation.
3. Cross-attention.

9. Next-token prediction-The model learns to predict the next correct token.